To execute this lab experiment, install **NLTK** and the required corpora, then run the code.

Step 1: Install NLTK

Open Command Prompt/Terminal:

```
pip install nltk
```

Step 2: Download Required Datasets

Run Python and execute:

```
import nltk

nltk.download('gutenberg')
nltk.download('wordnet')
nltk.download('omw-1.4')
```

Or create a file setup.py:

```
import nltk

nltk.download('gutenberg')
nltk.download('wordnet')
nltk.download('omw-1.4')
```

Run:

```
python setup.py
```

Step 3: Save the Experiment

Create a file named paradigmatic_syntagmatic.py:

```
import nltk
from nltk.collocations import (
    BigramCollocationFinder,
    BigramAssocMeasures
)
from nltk.corpus import gutenberg, wordnet as wn

# Syntagmatic: which words keep company?
words = [
     w.lower()
     for w in gutenberg.words("austen-emma.txt")
     if w.isalpha()
```

```
]

f = BigramCollocationFinder.from_words(words)

f.apply_freq_filter(5)

print("Top 10 PMI Bigrams")
print("-" * 40)

for bg in f.nbest(BigramAssocMeasures.pmi, 10):
    print(bg)

print("\nWordNet Synsets and Hypernyms")
print("-" * 40)

# Paradigmatic: which words can replace each other?
for syn in wn.synsets("car")[:3]:
    print("\nSynset:", syn.name())
    print("Lemmas:", syn.lemma_names())

    print("Hypernyms:")
    for h in syn.hypernyms():
        print("   ", h.name())
```

Step 4: Run the Program

```
python paradigmatic_syntagmatic.py
```

Expected Output

Syntagmatic Relations (Word Associations)

Example:

```
Top 10 PMI Bigrams
----------------------------------------
('sore', 'throat')
('baked', 'apples')
('maple', 'grove')
('weston', 'house')
('highbury', 'people')
```

```
8  ...
```

Interpretation

Words such as:

```
1  ('sore', 'throat')
```

occur together frequently and therefore have a strong **syntagmatic relationship**.

Syntagmatic relations answer:

> "Which words tend to occur together in a sentence?"

Examples:

- sore → throat
- baked → apples
- ice → cream
- coffee → cup

Paradigmatic Relations (Substitution Relations)

Example:

```
1  Synset: car.n.01
2  Lemmas: ['car', 'auto', 'automobile', 'machine', 'motorcar']
3
4  Hypernyms:
5      motor_vehicle.n.01
```

Interpretation

Words in the same synset can often replace one another:

```
1  car
2  automobile
3  auto
4  motorcar
```

These are **paradigmatic relations**.

Paradigmatic relations answer:

> "Which words can substitute for each other in a similar context?"

Examples:

```
1  car ⇔ automobile
2  teacher ⇔ instructor
```

```
student ↔ pupil
purchase ↔ buy
```

Lab Result

Syntagmatic Analysis

Using the Gutenberg corpus (*Emma* by Jane Austen), the experiment identifies word pairs that frequently occur together. Examples include:

```
sore throat
baked apples
```

These demonstrate syntagmatic relationships because the words co-occur in meaningful contexts.

Paradigmatic Analysis

Using WordNet, synonyms and hypernyms of the word **car** are extracted.

Example:

```
car.n.01
→ automobile
→ auto
→ motorcar

Hypernym:
→ motor_vehicle.n.01
```

These demonstrate paradigmatic relationships because the words belong to the same semantic class and can often replace one another.

Viva Questions

1. What is a syntagmatic relation?

A syntagmatic relation is the relationship between words that commonly occur together in a sentence or phrase.

Example: sore throat

2. What is a paradigmatic relation?

A paradigmatic relation is the relationship between words that can substitute for one another.

Example: car and automobile

3. Which NLTK class finds collocations?

```
BigramCollocationFinder
```

4. Which corpus is used for semantic relations?

```
WordNet
```

5. What is a hypernym?

A hypernym is a more general category.

Example:

```
car → motor_vehicle
dog → animal
rose → flower
```

Conclusion

This experiment demonstrates two important linguistic relationships:

1. **Syntagmatic Relations**: words that occur together (e.g., *sore throat*).
2. **Paradigmatic Relations**: words that can replace one another (e.g., *car* and *automobile*).

These concepts are fundamental in **Natural Language Processing (NLP)** for collocation extraction, semantic analysis, information retrieval, and language understanding.